SETK 4643

Thermodynamics

Chapter 5: Mass And Energy Analysis Of Control Volumes

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5.1 Conservation of Mass

- Conservation of mass principle is expressed as net mass transfer to or from a control volume during a time interval that is equal to the net change (increase or decrease) in the total mass within the control volume during the time interval.
- Mass, like energy, is a conserved property, and it cannot be created or destroyed during a process.
- Technically, mass *m* and energy *E* can be converted to each other according to the well-known formula proposed by Albert Einstein (1879-1955):

$$E = mc^2$$

where *c* is the speed of light in a vacuum, $c = 2.9979 \times 10^8$ m/s.

- The mass change due to energy change is negligible.

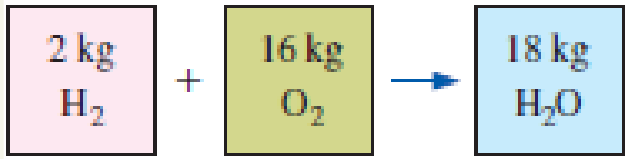


FIGURE 5–1
Mass is conserved even during chemical reactions.

5.1 Conservation of Mass (2)

- For **closed systems**, the conservation of mass principle is implicitly used by requiring that the mass of the system remain constant during a process.
- For **control volumes**, however, mass can cross the boundaries, and so we must keep track of the amount of mass entering and leaving the control volume.

5.1 Conservation of Mass (3)

Mass and Volume Flow Rates

- The amount of mass flowing through a cross section per unit time is called the mass flow rate and is denoted by $\dot{m}$.
- A fluid usually flows into or out of a control volume through pipes or ducts.
- The differential mass flow rate of fluid flowing across a small area element dA_c on a flow cross section is proportional to dA_c, the fluid density ρ, and the component of the flow velocity normal to dA_c, which we denote as V_n, and is expressed as

$$\delta\dot{m} = \rho V_n dA_c$$

- The mass flow rate through the entire cross-sectional area of a pipe or duct is obtained by integration:

$$\dot{m} = \int_{A_c} \delta\dot{m} = \int_{A_c} \rho V_n dA_c \qquad (kg/s)$$

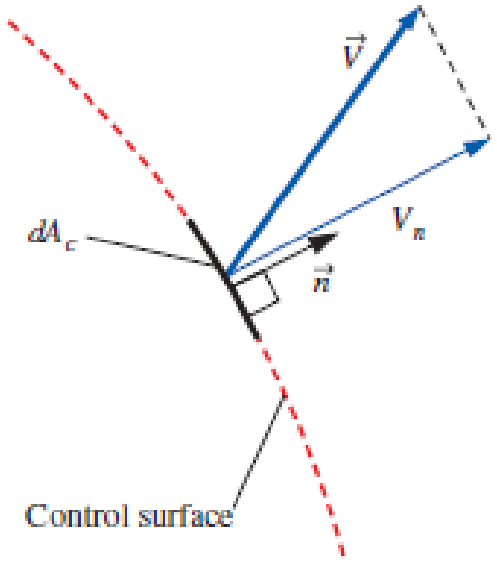


FIGURE 5–2
The normal velocity V_n for a surface is the component of velocity perpendicular to the surface.

5.1 Conservation of Mass (4)

- Velocity is never uniform over a cross section of a pipe. It varies from zero at the walls to some maximum value at or near the center line of the pipe.
- Average velocity V_{avg} is the average value of the normal velocity V_n across an entire flow cross section.

$$V_{avg} = \frac{1}{A_c} \int_{A_c} V_n \, dA_c$$

where A_c is the area of the cross section normal to the flow direction.

- For incompressible flow or compressible flow where ρ is uniform across A_c,

$$\dot{m} = \rho V_{avg} A_c \quad (kg/s)$$

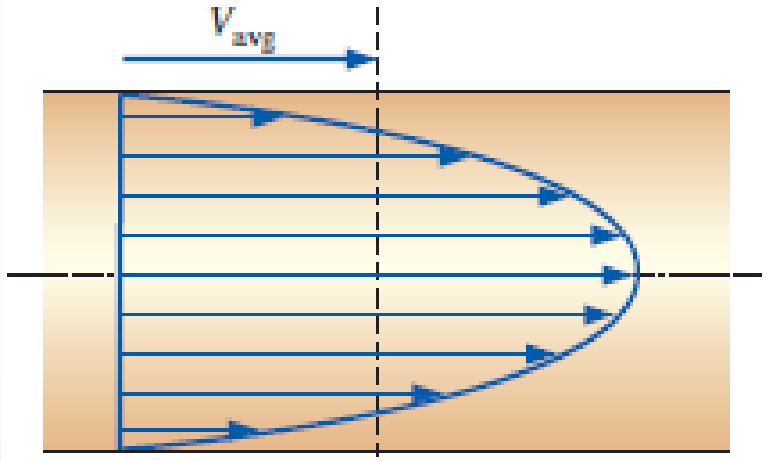


FIGURE 5–3
The average velocity V_{avg} is defined as the average speed through a cross section.

5.1 Conservation of Mass (4)

- The fluid volume flowing through a cross section per unit time is called the volume flow rate, $\dot{V}$ and is given by

$$\dot{V} = \int_{A_c} V_n \, dA_c = V_{avg} \, A_c = VA_c \qquad (m^3/s)$$

- The mass and volume flow rates are related by

$$\dot{m} = \rho \dot{V} = \frac{\dot{V}}{v}$$

where v is the specific volume.

- This relation is analogous to $m = \rho V = V/v$, which is the relation between the mass and the volume of a fluid in a container.

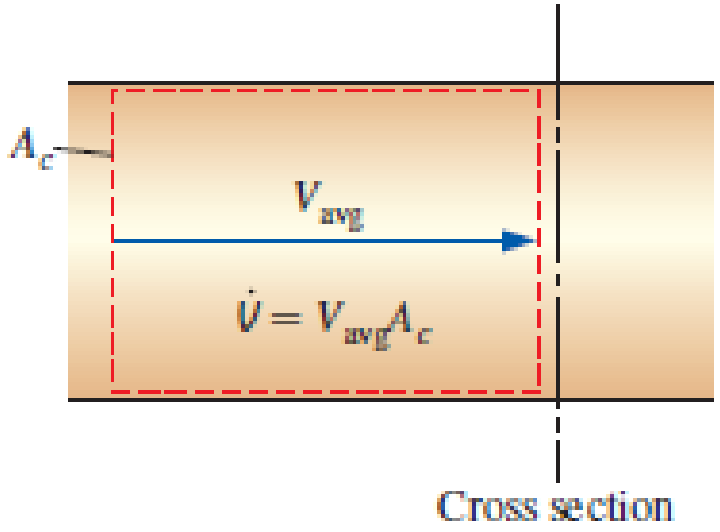


FIGURE 5–4
The volume flow rate is the volume of fluid flowing through a cross section per unit time.

5.1 Conservation of Mass (5)

Conservation of Mass Principle

- Conservation of mass principle is expressed as net mass transfer to or from a control volume during a time interval Δt is equal to the net change (increase or decrease) in the total mass within the control volume during Δt. That is,

$$\begin{pmatrix}\text{Total mass entering}\\ \text{the CV during } \Delta t\end{pmatrix} - \begin{pmatrix}\text{Total mass leaving}\\ \text{the CV during } \Delta t\end{pmatrix} = \begin{pmatrix}\text{Net change of mass}\\ \text{within the CV during } \Delta t\end{pmatrix}$$

or

$$m_{in} - m_{out} = \Delta m_{CV} \qquad \text{(kg)}$$

Where $\Delta m_{CV} = m_{final} - m_{initial}$ is the change in the mass of the control volume during the process.

- In rate form,

$$\dot{m}_{in} - \dot{m}_{out} = dm_{CV}/dt \qquad \text{(kg/s)}$$

5.1 Conservation of Mass (6)

- Total mass within the CV: $$m_{CV} = \int_{CV} \rho \, dV$$

- Then the time rate of change of the amount of mass within the control volume can be expressed as

$$\frac{dm_{CV}}{dt} = \frac{d}{dt}\int_{CV} \rho \, dV$$

Rate of change of mass within the CV

- The general conservation of mass relation can be expressed as

$$\boxed{\frac{d}{dt}\int_{CV} \rho \, dV = \sum_{in} \dot{m} - \sum_{out} \dot{m} \quad \text{or} \quad \frac{dm_{CV}}{dt} = \sum_{in} \dot{m} - \sum_{out} \dot{m}}$$

5.1 Conservation of Mass (7)

Mass Balance for Steady-Flow Processes

- During a steady-flow process, the total amount of mass contained within a control volume does not change with time (m_{CV} = constant).
- The conservation of mass principle for a general steady-flow system with multiple inlets and outlets can be expressed as (Fig. 5.3)

$$\sum_{in} \dot{m} = \sum_{out} \dot{m} \qquad \text{(kg/s)}$$

Steady flow

- It states that **the total rate of mass entering a control volume is equal to the total rate of mass leaving it**.
- Engineering devices such as nozzles, diffusers, turbines, compressors, and pumps involve a single stream (only one inlet and one outlet).
- For single stream steady flow systems:

$$\dot{m}_1 = \dot{m}_2 \quad \rightarrow \quad \rho_1 V_1 A_1 = \rho_2 V_2 A_2$$

Steady flow (single stream)

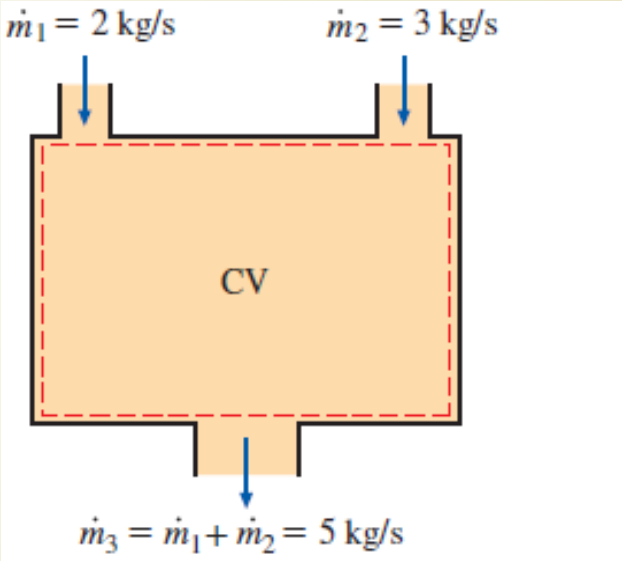


FIGURE 5–8
Conservation of mass principle for a two-inlet–one-outlet steady-flow system.

5.1 Conservation of Mass (7)

Special Case: Incompressible Flow

- For incompressible fluid (usually the case for liquids), the conservation of mass relations can be further simplified considering ρ = constant.

$$\sum_{in} \dot{V} = \sum_{out} \dot{V} \quad (m^3/s)$$

Steady, incompressible flow

- For single-stream steady-flow systems it becomes

$$\dot{V}_1 = \dot{V}_2 \rightarrow V_1 A_1 = V_2 A_2$$

Steady, incompressible flow (single stream)

- There is no such thing as a **conservation of volume** principle.
  - The volume flow rates into and out of a steady-flow device may be different.
  - The volume flow rate at the outlet of an air compressor is much less than that at the inlet even though the mass flow rate of air through the compressor is constant.
    - This is due to the higher density of air at the compressor exit.
    - What about liquids?

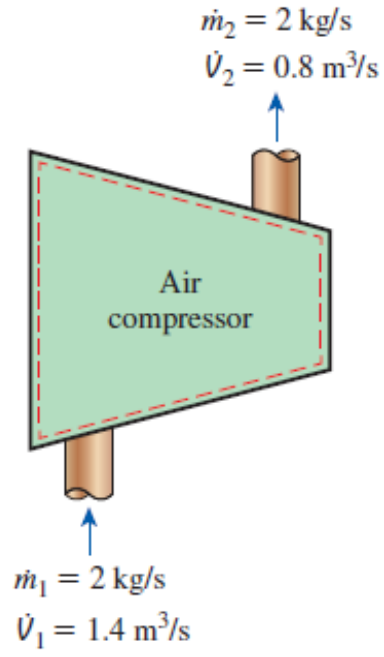


FIGURE 5–9
During a steady-flow process, volume flow rates are not necessarily conserved, although mass flow rates are.

EXAMPLE 5–1 Water Flow Through a Garden Hose Nozzle

A garden hose attached with a nozzle is used to fill a 10-gal bucket. The inner diameter of the hose is 2 cm, and it reduces to 0.8 cm at the nozzle exit (Fig. 5–10). If it takes 50 s to fill the bucket with water, determine (*a*) the volume and mass flow rates of water through the hose, and (*b*) the average velocity of water at the nozzle exit.

SOLUTION A garden hose is used to fill a water bucket. The volume and mass flow rates of water and the exit velocity are to be determined.

Assumptions **1** Water is a nearly incompressible substance. **2** Flow through the hose is steady. **3** There is no waste of water by splashing.

Properties We take the density of water to be 1000 kg/m^3 = 1 kg/L.

Analysis (*a*) Noting that 10 gal of water are discharged in 50 s, the volume and mass flow rates of water are

$$\dot{V} = \frac{V}{\Delta t} = \frac{10\text{ gal}}{50\text{ s}}\left(\frac{3.7854\text{ L}}{1\text{ gal}}\right) = \mathbf{0.757\ L/s}$$

$$\dot{m} = \rho\dot{V} = (1\text{ kg/L})(0.757\text{ L/s}) = \mathbf{0.757\ L/s}$$

(*b*) The cross-sectional area of the nozzle exit is

$$A_e = \pi r_e^2 = \pi(0.4\text{ cm})^2 = 0.5027\text{ cm}^2 = 0.5027 \times 10^{-4}\text{ m}^2$$

FIGURE 5–10
Schematic for Example 5–1.

The volume flow rate through the hose and the nozzle is constant. Then the average velocity of water at the nozzle exit becomes

$$V_e = \frac{\dot{V}}{A_e} = \frac{0.757\text{ L/s}}{0.5027 \times 10^{-4}\text{ m}^2}\left(\frac{1\text{ m}^3}{1000\text{ L}}\right) = \mathbf{15.1\ m/s}$$

Discussion It can be shown that the average velocity in the hose is 2.4 m/s. Therefore, the nozzle increases the water velocity by over six times.

5.2 Flow Work and the Energy of a Flowing Fluid

- Control volumes involve mass flow across their boundaries, and some work is required to push the mass into or out of the *CV*.
- This work is known as the **flow work**, or **flow energy**, and is necessary for maintaining a continuous flow through a *CV*.
- Consider a fluid element of volume *V*.
  - The fluid immediately upstream forces this fluid element to enter the *CV*; thus, can be regarded as an imaginary piston.
  - If the fluid pressure is *P* and the cross-sectional area of the fluid element is *A*, the force applied on the fluid element by the imaginary piston is

$$F = PA$$

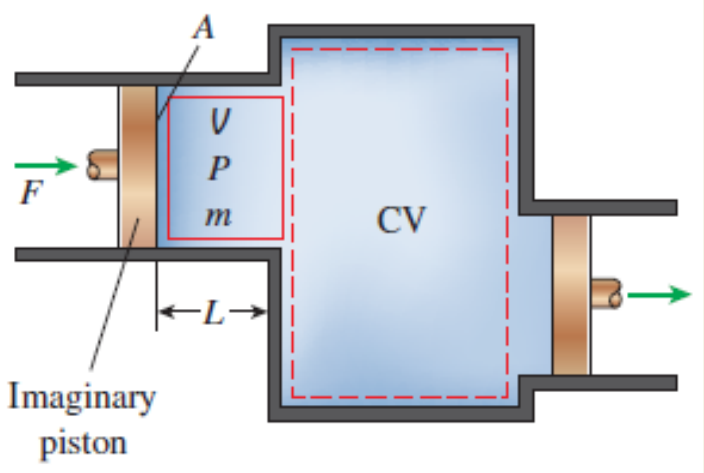


FIGURE 5–12
Schematic for flow work.

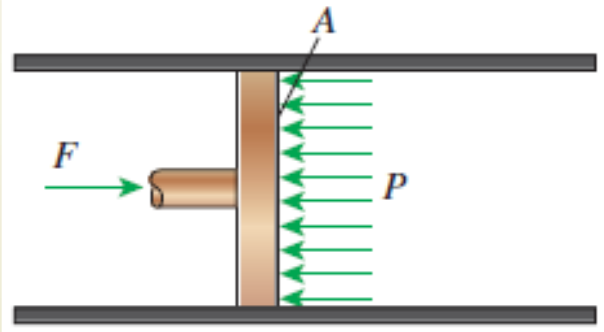


FIGURE 5–13
In the absence of acceleration, the force applied on a fluid by a piston is equal to the force applied on the piston by the fluid.

5.2 Flow Work and the Energy of a Flowing Fluid (2)

- To push the entire fluid element into the CV, this force must act through a distance *L*.
- Thus, the work done in pushing the fluid element across the boundary (i.e., the flow work) is

$$W_{\text{flow}} = FL = PAL = PV \qquad \text{(kJ)}$$

- The flow work per unit mass is obtained by dividing both sides of this equation by the mass of the fluid element:

$$w_{\text{flow}} = Pv \qquad \text{(kJ/kg)}$$

- Unlike other work quantities, flow work is expressed in terms of properties.
- In fact, it is the product of two properties of the fluid.

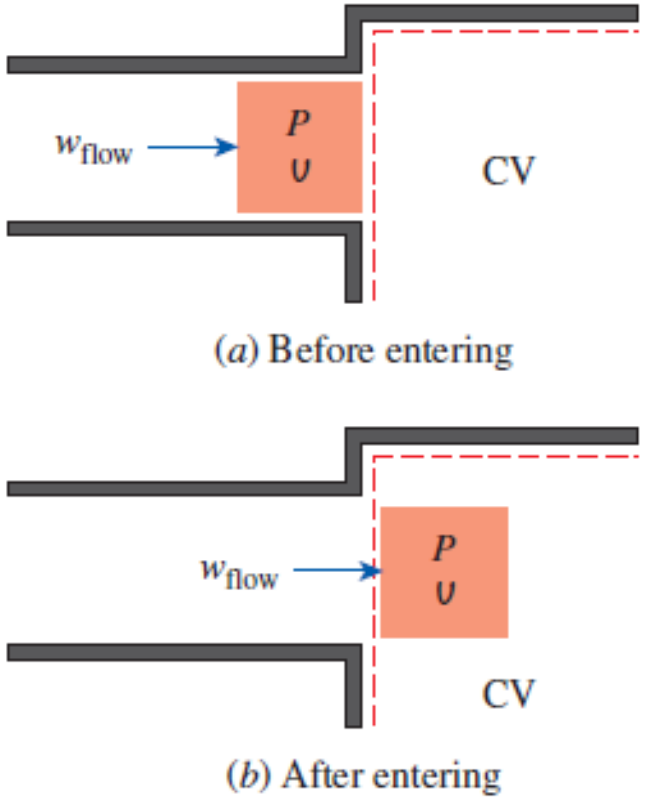


FIGURE 5–14
Flow work is the energy needed to push a fluid into or out of a control volume, and it is equal to Pv.

5.2 Flow Work and the Energy of a Flowing Fluid (3)

Total Energy of a Flowing Fluid

- The total energy of a simple compressible system consists of three parts: internal, kinetic, and potential energies.
- On a unit mass basis, it is expressed as

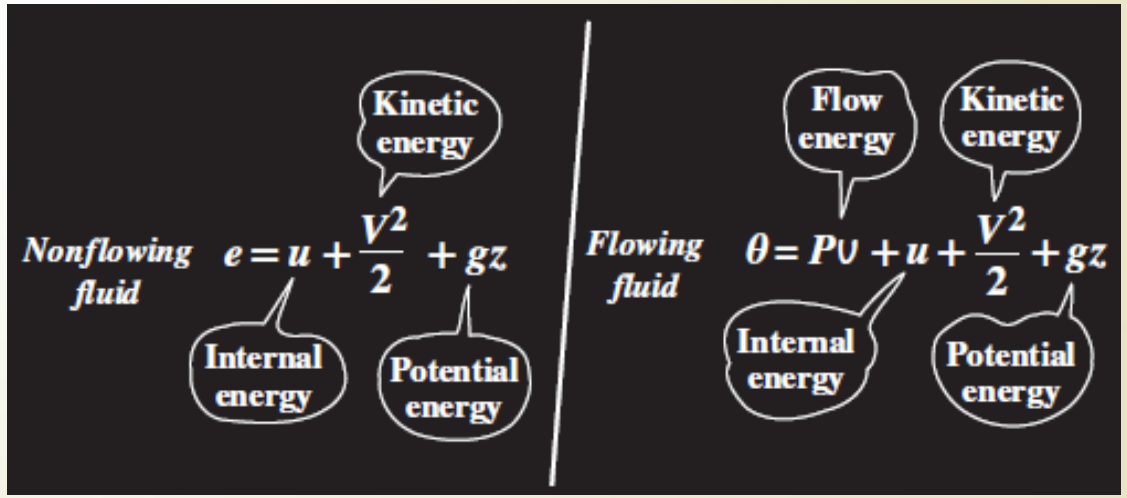
$$e = u + \text{ke} + \text{pe} = u + \frac{V^2}{2} + gz \qquad \text{(kJ/kg)}$$

- The fluid entering or leaving a CV possesses an additional form of energy – the **flow energy** *Pv*.
- The total energy of a flowing fluid on a unit-mass basis (denoted by θ) becomes

$$\theta = Pv + e = Pv + (u + \text{ke} + \text{pe})$$

or

$$\theta = h + \text{ke} + \text{pe} = h + \frac{V^2}{2} + gz \qquad \text{(kJ/kg)}$$



5.2 Flow Work and the Energy of a Flowing Fluid (4)

Energy Transport by Mass

- The total energy of a flowing fluid of mass m is simply mθ. When a fluid stream with uniform properties is flowing at a mass flow rate of $\dot{m}$, the rate of energy flow with that stream is $\dot{m}\theta$. That is,

$$E_{\text{mass}} = m\theta = m\left(h + \frac{V^2}{2} + gz\right) \quad \text{(kJ)}$$

Amount of energy transport by mass

$$\dot{E}_{\text{mass}} = \dot{m}\theta = \dot{m}\left(h + \frac{V^2}{2} + gz\right) \quad \text{(kW)}$$

Rate of energy transport by mass

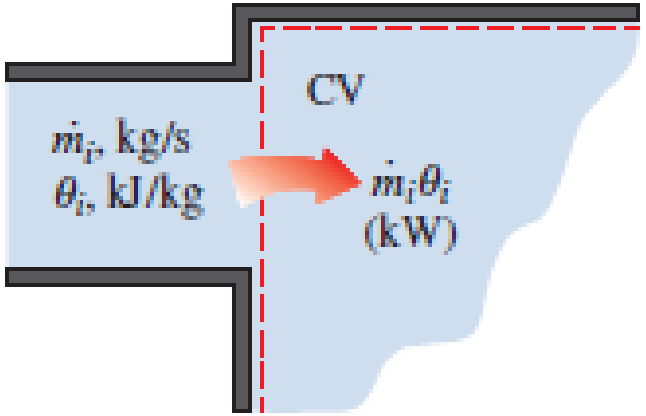


FIGURE 5–16
The product $\dot{m}_i\theta_i$ is the energy transported into control volume by mass per unit time.

EXAMPLE 5–3 Energy Transport by Flowing Air

Air flows steadily in a pipe at 300 kPa, 77°C, and 25 m/s at a rate of 18 kg/min (Fig. 5–17). Determine (*a*) the diameter of the pipe, (*b*) the rate of flow energy, (*c*) the rate of energy transport by mass, and (*d*) the error involved in part *c* if the kinetic energy is neglected.

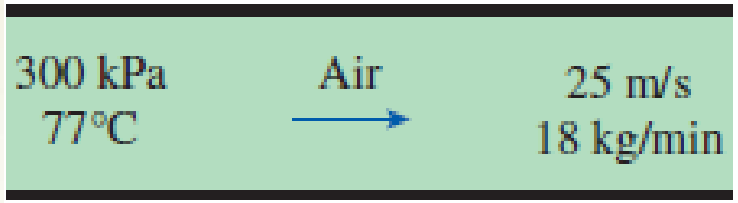


FIGURE 5–17
Schematic for Example 5–3.

SOLUTION Air flows steadily in a pipe at a specified state. The diameter of the pipe, the rate of flow energy, and the rate of energy transport by mass are to be determined. Also, the error involved in the determination of energy transport by mass is to be determined.

Assumptions **1** The flow is steady. **2** The potential energy is negligible.

Properties The properties of air are $R = 0.287$ kJ/kg·K and $c_p = 1.008$ kJ/kg·K (at 350 K from Table A-2*b*).

Analysis (*a*) The diameter is determined as follows:

$$v = \frac{RT}{P} = \frac{(0.287\ \text{kJ/kg·K})(77 + 273\ \text{K})}{300\ \text{kPa}} = 0.3349\ \text{m}^3/\text{kg}$$

$$A = \frac{\dot{m}v}{V} = \frac{(18/60\ \text{kg/s})(0.3349\ \text{m}^3/\text{kg})}{25\ \text{m/s}} = 0.004018\ \text{m}^2$$

$$D = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(0.004018\ \text{m}^2)}{\pi}} = \mathbf{0.0715\ m}$$

(*b*) The rate of flow energy is determined from

$$\dot{W}_{\text{flow}} = \dot{m}Pv = (18/60\ \text{kg/s})(300\ \text{kPa})(0.3349\ \text{m}^3/\text{kg}) = \mathbf{30.14\ kW}$$

(*c*) The rate of energy transport by mass is

$$\dot{E}_{\text{mass}} = \dot{m}(h + \text{ke}) = \dot{m}\left(c_p T + \frac{1}{2}V^2\right)$$

$$= (18/60\ \text{kg/s})\left[(1.008\ \text{kJ/kg·K})(77 + 273\ \text{K}) + \frac{1}{2}(25\ \text{m/s})^2\left(\frac{1\ \text{kJ/kg}}{1000\ \text{m}^2/\text{s}^2}\right)\right]$$

$$= \mathbf{105.94\ kW}$$

(*d*) If we neglect kinetic energy in the calculation of energy transport by mass

$$\dot{E}_{\text{mass}} = \dot{m}h = \dot{m}c_p T = (18/60\ \text{kg/s})(1.005\ \text{kJ/kg·K})(77 + 273\ \text{K}) = 105.84\ \text{kW}$$

Therefore, the error involved if we neglect the kinetic energy is only **0.09 percent.**

Discussion The numerical value of the energy transport with air alone does not mean much since this value depends on the reference point selected for enthalpy (it could even be negative). The significant quantity is the difference between the enthalpies of the air in the pipe and the ambient air since it relates directly to the amount of energy supplied to heat air from ambient temperature to 77°C.

Exercise 1

5–9 A 2-m^3 rigid tank initially contains air whose density is 1.18 kg/m^3. The tank is connected to a high-pressure supply line through a valve. The valve is opened, and air is allowed to enter the tank until the density in the tank rises to 5.30 kg/m^3. Determine the mass of air that has entered the tank. *Answer:* 8.24 kg

Exercise 2

5–21 Refrigerant-134a enters the compressor of a refrigeration system as saturated vapor at 0.14 MPa, and leaves as superheated vapor at 0.8 MPa and 60°C at a rate of 0.06 kg/s. Determine the rates of energy transfers by mass into and out of the compressor. Assume the kinetic and potential energies to be negligible.

5.3 Energy Analysis of Steady-Flow Systems

- (**Recall**) Steady-flow process is a process during which a fluid flows through a control volume steadily.
  - That is , the fluid properties can change from point to point within the CV, but at any point, they remain constant during the entire process.
- During a steady-flow process, no intensive or extensive properties within the control volume change with time.
- Thus, volume *V*, the mass *m*, and the total energy content *E* of the control volume remain constant.
- As a result,
  - the boundary work is zero for steady-flow systems (since V_{CV} = const.), and
  - the total mass or energy entering the CV must be equal to the total mass or energy leaving it (since m_{CV} = constant and E_{CV} = constant).

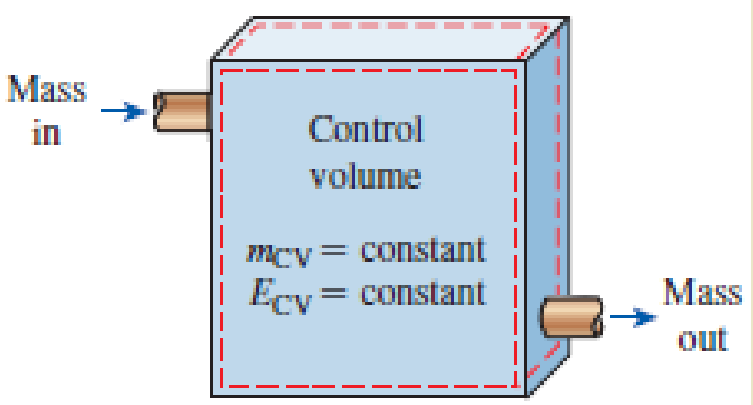


FIGURE 5–19
Under steady-flow conditions, the mass and energy contents of a control volume remain constant.

5.3 Energy Analysis of Steady-Flow Systems (2)

- The fluid properties at an inlet or exit remain constant during a steady-flow process.
  - The properties may, however, be different at different inlets and exits.
  - The mass flow rate of the fluid at an opening must remain constant during a steady-flow process.
  - The heat and work interactions between a steady-flow system and its surroundings do not change with time.
- The mass balance for a general steady-flow system:

$$\sum_{in} \dot{m} = \sum_{out} \dot{m} \qquad \text{(kg/s)}$$

- The mass balance for a single-stream (one-inlet and one-outlet) steady-flow system:

$$\dot{m}_1 = \dot{m}_2 \longrightarrow \rho_1 V_1 A_1 = \rho_2 V_2 A_2$$

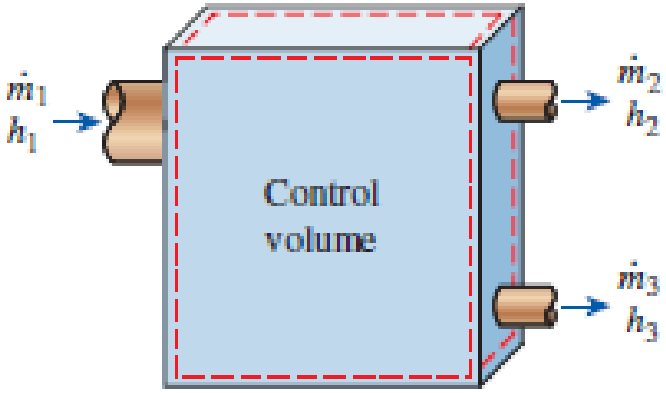


FIGURE 5–20
Under steady-flow conditions, the fluid properties at an inlet or exit remain constant (do not change with time).

5.3 Energy Analysis of Steady-Flow Systems (3)

- The rate form of the general energy balance for a steady-flow process is

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{dE_{system}/dt}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} = 0$$

0 (steady)

or

$$\textit{Energy balance:} \quad \underbrace{\dot{E}_{in}}_{\text{Rate of net energy transfer in by heat, work, and mass}} = \underbrace{\dot{E}_{out}}_{\text{Rate of net energy transfer out by heat, work, and mass}} \quad (\text{kW})$$

- Energy can be transferred by heat, work, and mass only, thus:

$$\dot{Q}_{in} + \dot{W}_{in} + \sum_{in} \dot{m}\theta = \dot{Q}_{out} + \dot{W}_{out} + \sum_{out} \dot{m}\theta$$

or

$$\dot{Q}_{in} + \dot{W}_{in} + \sum_{in} \underbrace{\dot{m}\left(h + \frac{V^2}{2} + gz\right)}_{\text{for each inlet}} = \dot{Q}_{out} + \dot{W}_{out} + \sum_{out} \underbrace{\dot{m}\left(h + \frac{V^2}{2} + gz\right)}_{\text{for each exit}}$$

5.3 Energy Analysis of Steady-Flow Systems (4)

- It is common practice to assume heat to be transferred into the system (heat input) at a rate of $\dot{Q}$, and work produced by the system (work output) at a rate of $\dot{W}$.
- The first-law of energy balance relation in that case for a general steady-flow system becomes

$$\dot{Q} - \dot{W} = \sum_{\text{out}} \underbrace{\dot{m}\left(h + \frac{V^2}{2} + gz\right)}_{\text{for each exit}} - - \sum_{\text{in}} \underbrace{\dot{m}\left(h + \frac{V^2}{2} + gz\right)}_{\text{for each inlet}}$$

- For single-stream devices, the steady-flow energy balance equation becomes

$$\dot{Q} - \dot{W} = \dot{m}\left[h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)\right]$$

$$q - w = h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)$$

5.4 Some Steady-Flow Engineering Devices

- Many engineering devices operate essentially under the same conditions for long periods of time.
  - The components of a steam power plant (turbines, compressors, heat exchangers, and pumps), for example, operate nonstop for months before the system is shut down for maintenance.
- Therefore, these devices can be conveniently analyzed as steady-flow devices.
- Some common steady-flow devices are described, and the thermodynamic aspects of the flow through them are analyzed.

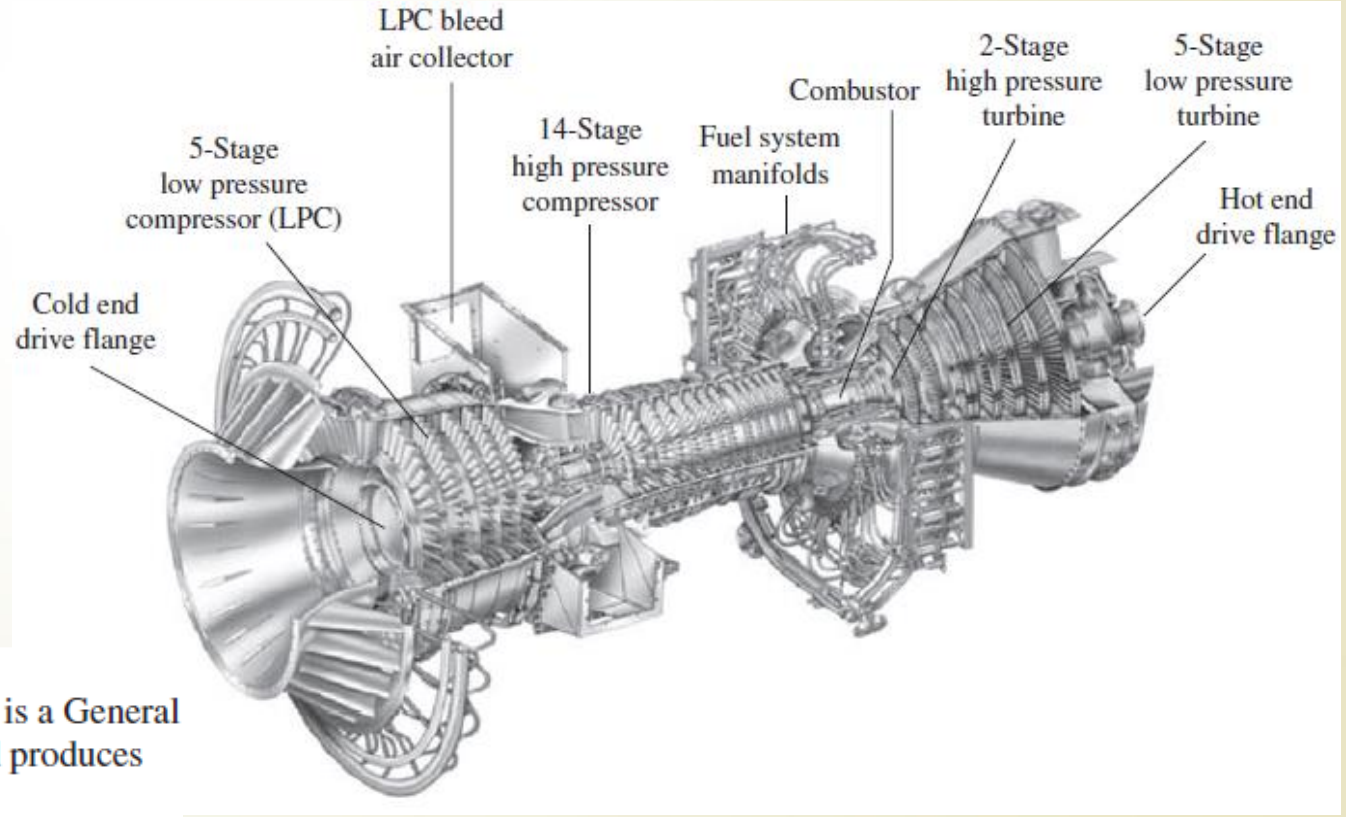


FIGURE 5–25
A modern land-based gas turbine used for electric power production. This is a General Electric LM5000 turbine. It has a length of 6.2 m, it weighs 12.5 tons, and produces 55.2 MW at 3600 rpm with steam injection.
Courtesy of GE Power Systems.

5.4 Some Steady-Flow Engineering Devices (2)

1. Nozzles and Diffusers

- A nozzle is a device that increases the velocity of a fluid at the expense of decreasing pressure.
- A diffuser is a device that increases the pressure of a fluid by decreasing the fluid velocity.
- Nozzles and diffusers perform opposite tasks.
- The rate of heat transfer between the fluid flowing through a nozzle or a diffuser and the surroundings is usually very small $(\dot{Q} \approx 0)$ since the fluid has high velocities, and thus it does not spend enough time in the device for any significant heat transfer to take place.

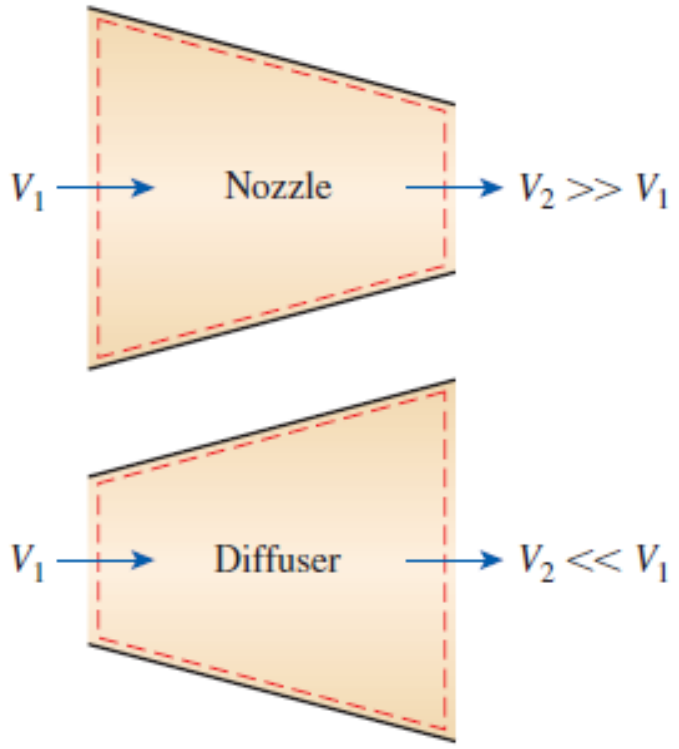


FIGURE 5–26
Nozzles and diffusers are shaped so that they cause large changes in fluid velocities and thus kinetic energies.

5.4 Some Steady-Flow Engineering Devices (3)

- Nozzles and diffusers typically involve no work $\left(\dot{W} \approx 0\right)$ and any change in potential energy is negligible ($\Delta pe \cong 0$).
- But nozzles and diffusers usually involve very high velocities, and as a fluid passes through a nozzle or diffuser, it experiences large changes in its velocity.
- Therefore, the kinetic energy changes must be accounted for in analyzing the flow through these devices ($\Delta ke \neq 0$).

EXAMPLE 5–4 Deceleration of Air in a Diffuser

Air at 10°C and 80 kPa enters the diffuser of a jet engine steadily with a velocity of 200 m/s. The inlet area of the diffuser is 0.4 m². The air leaves the diffuser with a velocity that is very small compared with the inlet velocity. Determine (*a*) the mass flow rate of the air and (*b*) the temperature of the air leaving the diffuser.

SOLUTION Air enters the diffuser of a jet engine steadily at a specified velocity. The mass flow rate of air and the temperature at the diffuser exit are to be determined.

Assumptions **1** This is a steady-flow process since there is no change with time at any point and thus $\Delta m_{CV} = 0$ and $\Delta E_{CV} = 0$. **2** Air is an ideal gas since it is at a high temperature and low pressure relative to its critical-point values. **3** The potential energy change is zero, $\Delta pe = 0$. **4** Heat transfer is negligible. **5** Kinetic energy at the diffuser exit is negligible. **6** There are no work interactions.

Analysis We take the *diffuser* as the system (Fig. 5–27). This is a *control volume* since mass crosses the system boundary during the process. We observe that there is only one inlet and one exit and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$.

(*a*) To determine the mass flow rate, we need to find the specific volume of the air first. This is determined from the ideal-gas relation at the inlet conditions:

$$v_1 = \frac{RT_1}{P_1} = \frac{(0.287\ \text{kPa·m}^3/\text{kg·K})(283\ \text{K})}{80\ \text{kPa}} = 1.015\ \text{m}^3/\text{kg}$$

FIGURE 5–27
The diffuser of a jet engine discussed in Example 5–4.
Photo by Yunus Çengel

Then,

$$\dot{m} = \frac{1}{v_1} V_1 A_1 = \frac{1}{1.015\ \text{m}^3/\text{kg}}(200\ \text{m/s})(0.4\ \text{m}^2) = \mathbf{78.8\ kg/s}$$

Since the flow is steady, the mass flow rate through the entire diffuser remains constant at this value.

(*b*) Under stated assumptions and observations, the energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\substack{\text{Rate of net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{dE_{system}/dt^{\,0\ (\text{steady})}}_{\substack{\text{Rate of change in internal, kinetic,} \\ \text{potential, etc., energies}}} = 0$$

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{m}\left(h_1 + \frac{V_1^2}{2}\right) = \dot{m}\left(h_2 + \frac{V_2^2}{2}\right) \quad (\text{since } \dot{Q} \cong 0,\ \dot{W} = 0, \text{ and } \Delta pe \cong 0)$$

$$h_2 = h_1 - \frac{V_2^2 - V_1^2}{2}$$

The exit velocity of a diffuser is usually small compared with the inlet velocity ($V_2 << V_1$); thus, the kinetic energy at the exit can be neglected. The enthalpy of air at the diffuser inlet is determined from the air table (Table A–17) to be

$$h_1 = h_{@\ 283\ \text{K}} = 283.14\ \text{kJ/kg}$$

Substituting, we get

$$h_2 = 283.14\ \text{kJ/kg} - \frac{0 - (200\ \text{m/s})^2}{2}\left(\frac{1\ \text{kJ/kg}}{1000\ \text{m}^2/\text{s}^2}\right)$$

$$= 303.14\ \text{kJ/kg}$$

From Table A–17, the temperature corresponding to this enthalpy value is

$$T_2 = \mathbf{303\ K}$$

Discussion This result shows that the temperature of the air increases by about 20°C as it is slowed down in the diffuser. The temperature rise of the air is mainly due to the conversion of kinetic energy to internal energy.

5.4 Some Steady-Flow Engineering Devices (4)

2. Turbines and Compressors

- Turbine is a device that produces shaft work due to a decrease of enthalpy, kinetic, and potential energies of a flowing fluid.
- Compressors, pumps and fans are devices used to increase the pressure of a fluid. Work is supplied to these devices from an external source through a rotating shaft.
  - A fan increases the pressure of a gas slightly, while compressor is capable of compressing the gas to very high pressure (typical pressure ratios > 3).
  - Pumps work very much like compressors except that they handle liquids instead of gases.

FIGURE 5–29
Turbine blades attached to the turbine shaft.
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5.4 Some Steady-Flow Engineering Devices (5)

- The turbines produce power output whereas compressors, pumps, and fans require power input.
- Heat transfer are usually negligible $\left(\dot{Q} \approx 0\right)$ since they are typically well insulated.
- Heat transfer is negligible for compressor unless there is intentional cooling
- Potential energy changes are negligible ($\Delta pe \cong 0$).
- The velocities involved in these devices, with the exception of turbines and fans, are usually too low ($\Delta ke \cong 0$).
- The fluid velocities encountered in most turbines are very high, and the fluid experiences a significant change in its kinetic energy.
  - However, this change is usually very small relative to the change in enthalpy, and thus it is often disregarded.

EXAMPLE 5–6 Compressing Air by a Compressor

Air at 100 kPa and 280 K is compressed steadily to 600 kPa and 400 K. The mass flow rate of the air is 0.02 kg/s, and a heat loss of 16 kJ/kg occurs during the process. Assuming the changes in kinetic and potential energies are negligible, determine the necessary power input to the compressor.

SOLUTION Air is compressed steadily by a compressor to a specified temperature and pressure. The power input to the compressor is to be determined.

Assumptions **1** This is a steady-flow process since there is no change with time at any point and thus $\Delta m_{CV} = 0$ and $\Delta E_{CV} = 0$. **2** Air is an ideal gas since it is at a high temperature and low pressure relative to its critical-point values. **3** The kinetic and potential energy changes are zero, $\Delta ke = \Delta pe = 0$.

Analysis We take the *compressor* as the system (Fig. 5–30). This is a *control volume* since mass crosses the system boundary during the process. We observe that there is only one inlet and one exit and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. Also, heat is lost from the system and work is supplied to the system.

Under stated assumptions and observations, the energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{dE_{system}/dt^{\,0\ (\text{steady})}}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} = 0$$

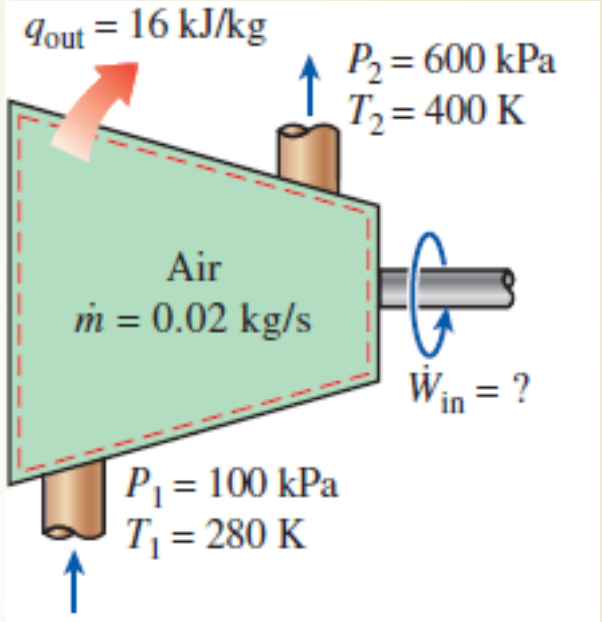


FIGURE 5–30
Schematic for Example 5–6.

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{W}_{in} + \dot{m}h_1 = \dot{Q}_{out} + \dot{m}h_2 \quad (\text{since } \Delta ke = \Delta pe \cong 0)$$

$$\dot{W}_{in} = \dot{m}q_{out} + \dot{m}(h_2 - h_1)$$

The enthalpy of an ideal gas depends on temperature only, and the enthalpies of the air at the specified temperatures are determined from the air table (Table A–17) to be

$$h_1 = h_{@\,280\,K} = 280.13 \text{ kJ/kg}$$

$$h_2 = h_{@\,400\,K} = 400.98 \text{ kJ/kg}$$

Substituting, the power input to the compressor is determined to be

$$\dot{W}_{in} = (0.02 \text{ kg/s})(16 \text{ kJ/kg}) + (0.02 \text{ kg/s})(400.98 - 280.13) \text{ kJ/kg}$$
$$= \mathbf{2.74 \text{ kW}}$$

Discussion Note that the mechanical energy input to the compressor manifests itself as a rise in enthalpy of air and heat loss from the compressor.

EXAMPLE 5–7 Power Generation by a Steam Turbine

The power output of an adiabatic steam turbine is 5 MW, and the inlet and the exit conditions of the steam are as indicated in Fig. 5–31.

(*a*) Compare the magnitudes of Δh, Δke, and Δpe.
(*b*) Determine the work done per unit mass of the steam flowing through the turbine.
(*c*) Calculate the mass flow rate of the steam.

SOLUTION The inlet and exit conditions of a steam turbine and its power output are given. The changes in kinetic energy, potential energy, and enthalpy of steam, as well as the work done per unit mass and the mass flow rate of steam are to be determined.

Assumptions **1** This is a steady-flow process since there is no change with time at any point and thus $\Delta m_{CV} = 0$ and $\Delta E_{CV} = 0$. **2** The system is adiabatic and thus there is no heat transfer.

Analysis We take the *turbine* as the system. This is a *control volume* since mass crosses the system boundary during the process. We observe that there is only one inlet and one exit and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. Also, work is done by the system. The inlet and exit velocities and elevations are given, and thus the kinetic and potential energies are to be considered.

(*a*) At the inlet, steam is in a superheated vapor state, and its enthalpy is

$$\left.\begin{aligned} P_1 &= 2\ \text{MPa} \\ T_1 &= 400°\text{C} \end{aligned}\right\} \quad h_1 = 3248.4\ \text{kJ/kg} \qquad \text{(Table A–6)}$$

At the turbine exit, we obviously have a saturated liquid–vapor mixture at 15-kPa pressure. The enthalpy at this state is

$$h_2 = h_f + x_2 h_{fg} = [225.94 + (0.9)(2372.3)]\ \text{kJ/kg} = 2361.01\ \text{kJ/kg}$$

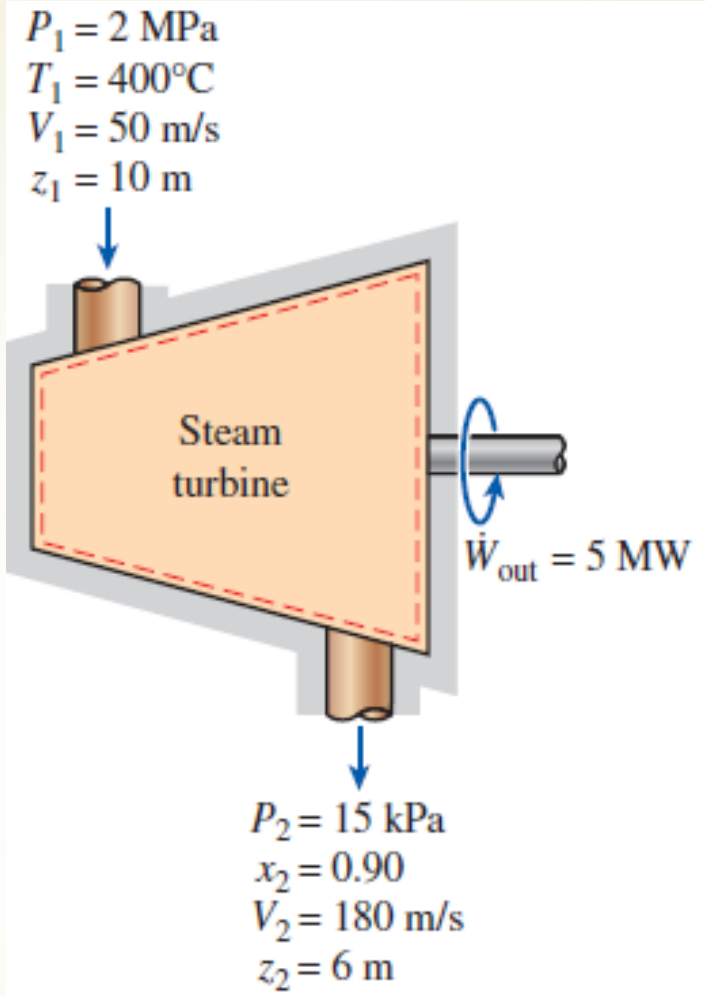


FIGURE 5–31
Schematic for Example 5–7.

Then

$$\Delta h = h_2 - h_1 = (2361.01 - 3248.4)\ \text{kJ/kg} = \mathbf{-887.39\ kJ/kg}$$

$$\Delta \text{ke} = \frac{V_2^2 - V_1^2}{2} = \frac{(180\ \text{m/s})^2 - (50\ \text{m/s})^2}{2}\left(\frac{1\ \text{kJ/kg}}{1000\ \text{m}^2/\text{s}^2}\right) = \mathbf{14.95\ kJ/kg}$$

$$\Delta \text{pe} = g(z_2 - z_1) = (9.81\ \text{m/s}^2)[(6 - 10)\ \text{m}]\left(\frac{1\ \text{kJ/kg}}{1000\ \text{m}^2/\text{s}^2}\right) = \mathbf{-0.04\ kJ/kg}$$

(*b*) The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\substack{\text{Rate of net energy transfer}\\ \text{by heat, work, and mass}}} = \underbrace{dE_{\text{system}}/dt}_{\substack{\text{Rate of change in internal, kinetic,}\\ \text{potential, etc., energies}}} = 0$$

0 (steady)

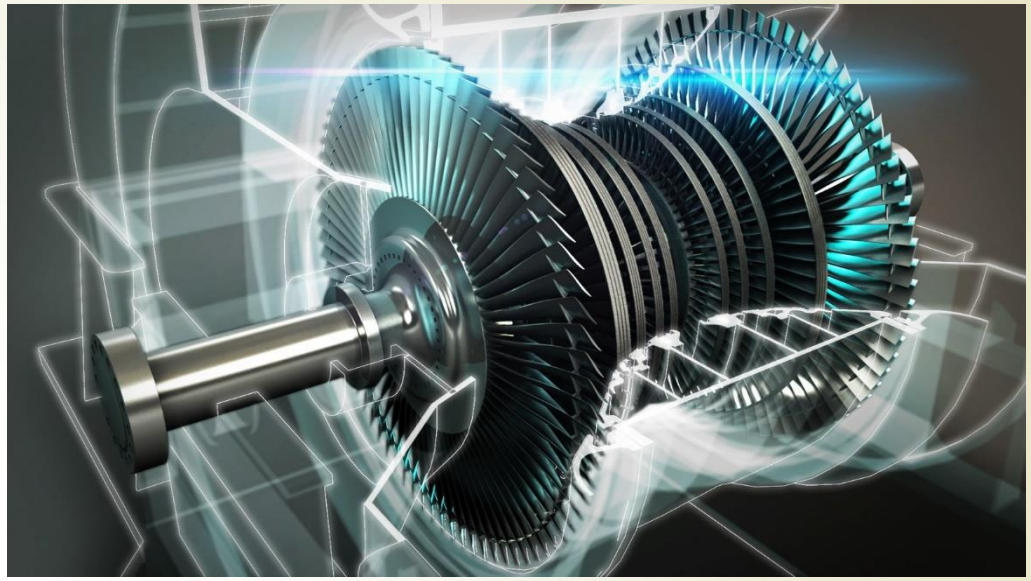
$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}\left(h_1 + \frac{V_1^2}{2} + gz_1\right) = \dot{W}_{\text{out}} + \dot{m}\left(h_2 + \frac{V_2^2}{2} + gz_2\right) \quad (\text{since } \dot{Q} = 0)$$

Dividing by the mass flow rate $\dot{m}$ and substituting, the work done by the turbine per unit mass of the steam is determined to be

$$w_{\text{out}} = -\left[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)\right] = -(\Delta h + \Delta \text{ke} + \Delta \text{pe})$$

$$= -[-887.39 + 14.95 - 0.04]\ \text{kJ/kg} = \mathbf{872.48\ kJ/kg}$$



(*c*) The required mass flow rate for a 5-MW power output is

$$\dot{m} = \frac{\dot{W}_{\text{out}}}{w_{\text{out}}} = \frac{5000\ \text{kJ/s}}{872.48\ \text{kJ/kg}} = \mathbf{5.73\ kg/s}$$

Discussion Two observations can be made from these results. First, the change in potential energy is insignificant in comparison to the changes in enthalpy and kinetic energy. This is typical for most engineering devices. Second, as a result of low pressure and thus high specific volume, the steam velocity at the turbine exit can be very high. Yet the change in kinetic energy is a small fraction of the change in enthalpy (less than 2 percent in our case) and is therefore often neglected.

5.4 Some Steady-Flow Engineering Devices (6)

3. Throttling Valves

- Throttling valves are any kind of flow-restricting devices that cause a significant pressure drop in a flowing fluid.
  - Some examples are ordinary adjustable valves, capillary tubes, and porous plugs.
- Unlike turbines, they produce a pressure drop without involving any work.
- The pressure drop in the fluid is often accompanied by a large drop in temperature.
  - Reason why throttling devices are commonly used in refrigeration and air-conditioning applications.
- Throttling valves are usually small devices.
  - the flow may be assumed adiabatic ($q \cong 0$) since there is neither sufficient time nor large enough area for any effective heat transfer to take place.
- Also there is no work done ($w = 0$), and the change in potential energy ($\Delta pe \cong 0$).
- The increase in kinetic energy is insignificant ($\Delta ke \cong 0$).

5.4 Some Steady-Flow Engineering Devices (7)

- The conservation of energy equation for this single-stream steady-flow device (throttling valve) reduces to

$$h_2 \cong h_1 \qquad \text{(kJ/kg)}$$

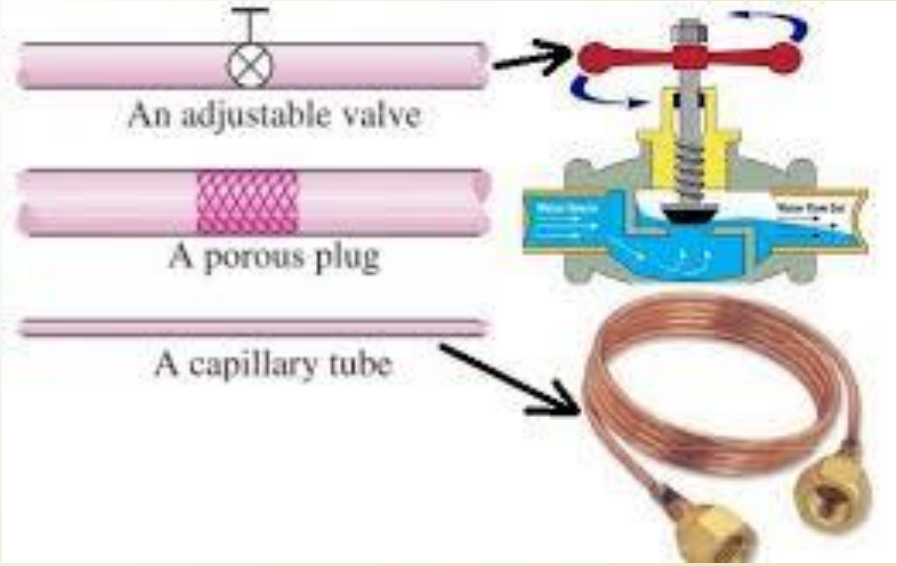
- Enthalpy values at the inlet and exit of a throttling valve are same.
- So, a throttling valve is sometimes called an isenthalpic device.

$$u_1 + P_1 v_1 = u_2 + P_2 v_2$$

or

$$\text{Internal energy} + \text{Flow energy} = \text{Constant}$$

- If the flow energy increases during the process ($P_2 v_2 > P_1 v_1$), it can do so at the expense of internal energy.
- As a result, internal energy decreases, which is usually accompanied by a drop in temperature.



5.4 Some Steady-Flow Engineering Devices (7)

- If the product *Pv* decreases, the internal energy and the temperature of a fluid will increase during a throttling process.
- In the case of an ideal gas, $h = h(T)$, and thus the temperature has to remain constant during a throttling process

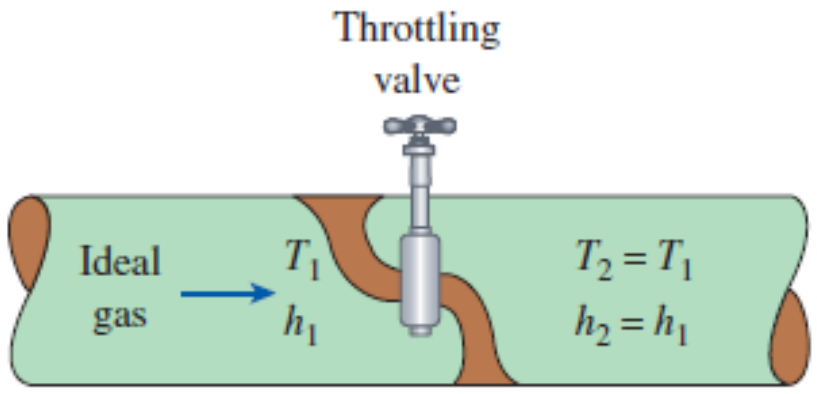


FIGURE 5–33
The temperature of an ideal gas does not change during a throttling (h = constant) process since $h = h(T)$.

EXAMPLE 5–8 Expansion of Refrigerant-134a in a Refrigerator

Refrigerant-134a enters the capillary tube of a refrigerator as saturated liquid at 0.8 MPa and is throttled to a pressure of 0.12 MPa. Determine the quality of the refrigerant at the final state and the temperature drop during this process.

SOLUTION Refrigerant-134a that enters a capillary tube as saturated liquid is throttled to a specified pressure. The exit quality of the refrigerant and the temperature drop are to be determined.

Assumptions **1** Heat transfer from the tube is negligible. **2** Kinetic energy change of the refrigerant is negligible.

Analysis A capillary tube is a simple flow-restricting device that is commonly used in refrigeration applications to cause a large pressure drop in the refrigerant. Flow through a capillary tube is a throttling process; thus, the enthalpy of the refrigerant remains constant (Fig. 5–34).

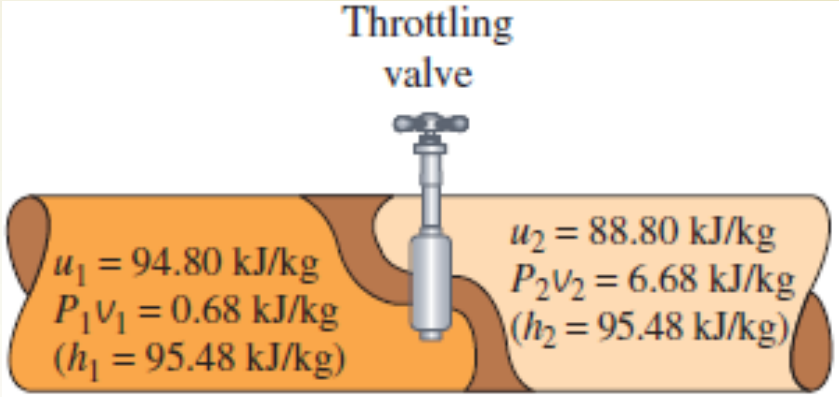


FIGURE 5–34
During a throttling process, the enthalpy (flow energy + internal energy) of a fluid remains constant. But internal and flow energies may be converted to each other.

$$\text{At inlet:}\quad \left.\begin{array}{l} P_1 = 0.8\ \text{MPa} \\ \text{sat. liquid} \end{array}\right\} \begin{array}{l} T_1 = T_{\text{sat @ 0.8 MPa}} = 31.31°\text{C} \\ h_1 = h_{f\,@\,0.8\ \text{MPa}} = 95.48\ \text{kJ/kg} \end{array} \quad \text{(Table A–12)}$$

$$\text{At exit:}\quad \begin{array}{l} P_2 = 0.12\ \text{MPa} \\ (h_2 = h_1) \end{array} \longrightarrow \begin{array}{l} h_f = 22.47\ \text{kJ/kg} \quad T_{\text{sat}} = -22.32°\text{C} \\ h_g = 236.99\ \text{kJ/kg} \end{array}$$

Obviously $h_f < h_2 < h_g$; thus, the refrigerant exists as a saturated mixture at the exit state. The quality at this state is

$$x_2 = \frac{h_2 - h_f}{h_{fg}} = \frac{95.48 - 22.47}{236.99 - 22.47} = \mathbf{0.340}$$

Since the exit state is a saturated mixture at 0.12 MPa, the exit temperature must be the saturation temperature at this pressure, which is −22.32°C. Then the temperature change for this process becomes

$$\Delta T = T_2 - T_1 = (-22.32 - 31.31)°\text{C} = \mathbf{-53.63°C}$$

Discussion Note that the temperature of the refrigerant drops by 53.63°C during this throttling process. Also note that 34.0 percent of the refrigerant vaporizes during this throttling process, and the energy needed to vaporize

refrigerant. Flow through a capillary tube is a throttling process; thus, the enthalpy of the refrigerant remains constant (Fig. 5–34).

$$\textit{At inlet:}\quad \left.\begin{array}{l} P_1 = 0.8\ \text{MPa} \\ \text{sat. liquid} \end{array}\right\} \begin{array}{l} T_1 = T_{\text{sat @ 0.8 MPa}} = 31.31°\text{C} \\ h_1 = h_{f\,\text{@ 0.8 MPa}} = 95.48\ \text{kJ/kg} \end{array} \quad \text{(Table A–12)}$$

$$\textit{At exit:}\quad \begin{array}{l} P_2 = 0.12\ \text{MPa} \\ (h_2 = h_1) \end{array} \longrightarrow \begin{array}{l} h_f = 22.47\ \text{kJ/kg} \\ h_g = 236.99\ \text{kJ/kg} \end{array} \quad T_{\text{sat}} = -22.32°\text{C}$$

Obviously $h_f < h_2 < h_g$; thus, the refrigerant exists as a saturated mixture at the exit state. The quality at this state is

$$x_2 = \frac{h_2 - h_f}{h_{fg}} = \frac{95.48 - 22.47}{236.99 - 22.47} = \mathbf{0.340}$$

Since the exit state is a saturated mixture at 0.12 MPa, the exit temperature must be the saturation temperature at this pressure, which is −22.32°C. Then the temperature change for this process becomes

$$\Delta T = T_2 - T_1 = (-22.32 - 31.31)°\text{C} = \mathbf{-53.63°C}$$

Discussion Note that the temperature of the refrigerant drops by 53.63°C during this throttling process. Also note that 34.0 percent of the refrigerant vaporizes during this throttling process, and the energy needed to vaporize this refrigerant is absorbed from the refrigerant itself.

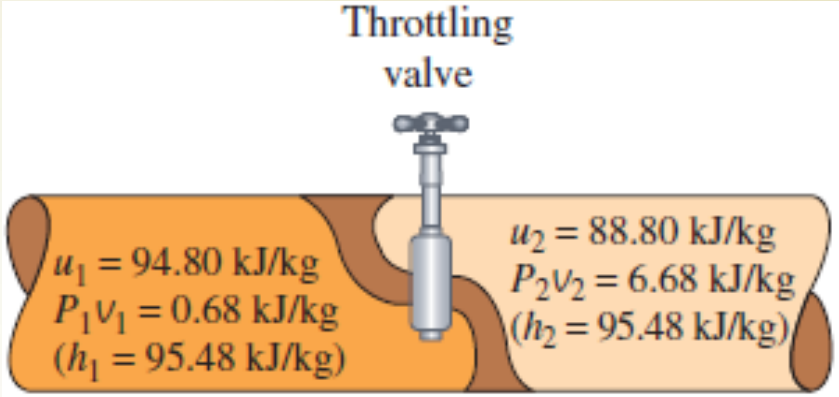


FIGURE 5–34
During a throttling process, the enthalpy (flow energy + internal energy) of a fluid remains constant. But internal and flow energies may be converted to each other.

5.4 Some Steady-Flow Engineering Devices (6)

4a. Mixing Chambers

- Mixing chamber is the section of a control volume where the mixing process takes place for two or more streams of fluids.
- An ordinary T-elbow or a Y-elbow in a shower, for example, serves as the mixing chamber for the cold- and hot-water streams
- Mixing chambers are usually well insulated ($q \cong 0$) and usually do not involve any kind of work ($w = 0$).
- Also, the kinetic and potential energies of the fluid streams are usually negligible (ke $\cong$ 0, pe $\cong$ 0)
- The conservation of mass principle:

$$\dot{m}_{in} = \dot{m}_{out} \rightarrow \dot{m}_1 + \dot{m}_2 = \dot{m}_3$$

- The conservation of energy principle:

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{m}_3 h_3 \quad (\text{since } \dot{Q} \cong 0, \dot{W} = 0, \text{ke} \cong \text{pe} \cong 0)$$

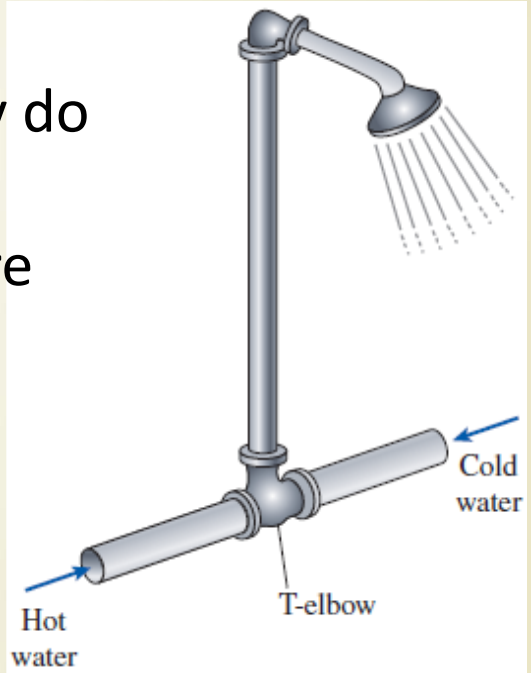


FIGURE 5–35
The T-elbow of an ordinary shower serves as the mixing chamber for the hot- and the cold-water streams.

5.4 Some Steady-Flow Engineering Devices (7)

4b. Heat Exchangers (HE)

- Heat exchangers are devices where two moving fluid streams exchange heat without mixing.
- The simplest form of a HE is double-tube (also called tube-and-shell)
  - HE composed of two concentric pipes of different diameters.
    - One fluid flows in the inner pipe, and the other in the annular space between the two pipes.
    - Heat is transferred from the hot fluid to the cold one through the wall separating them
    - Sometimes the inner tube makes a couple of turns inside the shell to increase the heat transfer area, and thus the rate of heat transfer.
- HEs typically involve no work interactions ($w = 0$) and negligible kinetic and potential energy changes ($ke \cong 0$, $pe \cong 0$) for each fluid stream.

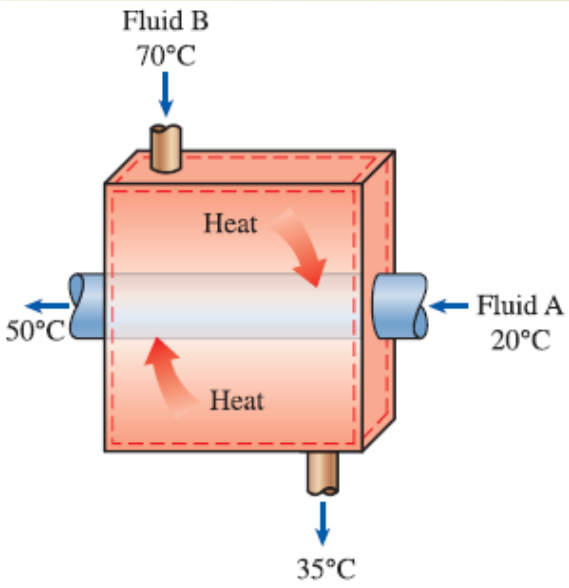
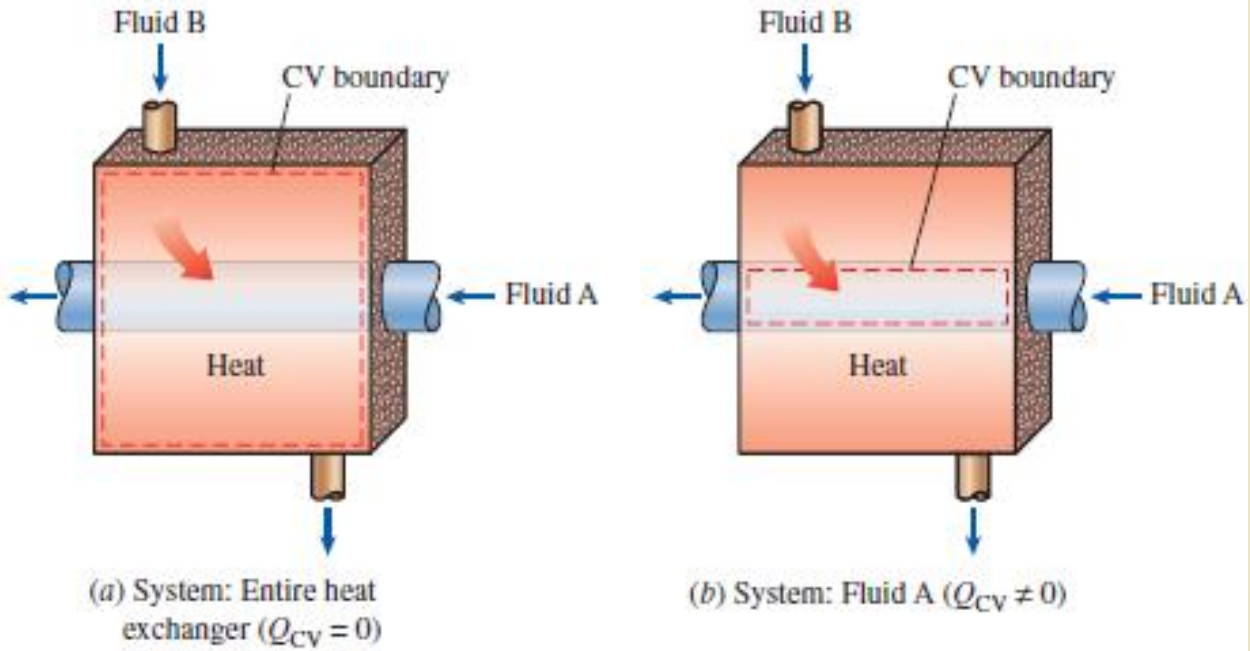


FIGURE 5–38
A heat exchanger can be as simple as two concentric pipes.

5.4 Some Steady-Flow Engineering Devices (8)

- The outer shell is usually well insulated to prevent any heat loss to the surrounding medium.
- The heat transfer rate associated with heat exchangers depends on how the control volume is selected.
  - When the entire heat exchanger is selected as the control volume, $\dot{Q} = 0$
  - If only one of the fluids is selected as the control volume, then heat will cross this boundary as it flows from one fluid to the other, and $\dot{Q} \neq 0$.



(a) System: Entire heat exchanger ($Q_{CV} = 0$)

(b) System: Fluid A ($Q_{CV} \neq 0$)

FIGURE 5–39
The heat transfer associated with a heat exchanger may be zero or nonzero depending on how the control volume is selected.

EXAMPLE 5–10 Cooling of Refrigerant-134a by Water

Refrigerant-134a is to be cooled by water in a condenser. The refrigerant enters the condenser with a mass flow rate of 6 kg/min at 1 MPa and 70°C and leaves at 35°C. The cooling water enters at 300 kPa and 15°C and leaves at 25°C. Neglecting any pressure drops, determine (*a*) the mass flow rate of the cooling water required and (*b*) the heat transfer rate from the refrigerant to water.

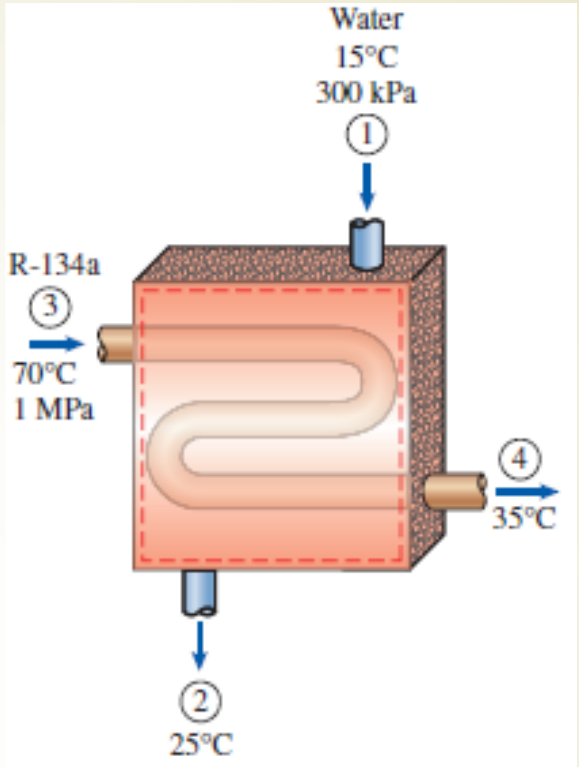


FIGURE 5–40
Schematic for Example 5–10.

SOLUTION Refrigerant-134a is cooled by water in a condenser. The mass flow rate of the cooling water and the rate of heat transfer from the refrigerant to the water are to be determined.

Assumptions **1** This is a steady-flow process since there is no change with time at any point and thus $\Delta m_{CV} = 0$ and $\Delta E_{CV} = 0$. **2** The kinetic and potential energies are negligible, ke ≅ pe ≅ 0. **3** Heat losses from the system are negligible and thus $\dot{Q} \cong 0$. **4** There is no work interaction.

Analysis We take the *entire heat exchanger* as the system (Fig. 5–40). This is a *control volume* since mass crosses the system boundary during the process. In general, there are several possibilities for selecting the control volume for multiple-stream steady-flow devices, and the proper choice depends on the situation at hand. We observe that there are two fluid streams (and thus two inlets and two exits) but no mixing. (*a*) Under the stated assumptions and observations, the mass and energy balances for this steady-flow system can be expressed in the rate form as follows:

Mass balance: $$\dot{m}_{in} = \dot{m}_{out}$$

for each fluid stream since there is no mixing. Thus,

$$\dot{m}_1 = \dot{m}_2 = \dot{m}_w$$

$$\dot{m}_3 = \dot{m}_4 = \dot{m}_R$$

Energy balance:

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{dE_{system}/dt}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} = 0$$

0 (steady)

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{m}_1 h_1 + \dot{m}_3 h_3 = \dot{m}_2 h_2 + \dot{m}_4 h_4 \quad (\text{since } \dot{Q} \cong 0,\ \dot{W} = 0,\ \text{ke} \cong \text{pe} \cong 0)$$

Combining the mass and energy balances and rearranging give

$$\dot{m}_w(h_1 - h_2) = \dot{m}_R(h_4 - h_3)$$

Now we need to determine the enthalpies at all four states. Water exists as a compressed liquid at both the inlet and the exit since the temperatures at both locations are below the saturation temperature of water at 300 kPa (133.52°C). Approximating the compressed liquid as a saturated liquid at the given temperatures, we have

$$h_1 \cong h_{f\,@\,15°C} = 62.982 \text{ kJ/kg}$$

$$h_2 \cong h_{f\,@\,25°C} = 104.83 \text{ kJ/kg}$$

(Table A–4)

The refrigerant enters the condenser as a superheated vapor and leaves as a compressed liquid at 35°C. From refrigerant-134a tables,

$$\left.\begin{aligned} P_3 &= 1 \text{ MPa} \\ T_3 &= 70°C \end{aligned}\right\} \quad h_3 = 303.87 \text{ kJ/kg} \qquad \text{(Table A–13)}$$

$$\left.\begin{aligned} P_4 &= 1 \text{ MPa} \\ T_4 &= 35°C \end{aligned}\right\} \quad h_4 \cong h_{f\,@\,35°C} = 100.88 \text{ kJ/kg} \qquad \text{(Table A–11)}$$

Substituting, we find

$$\dot{m}_w(62.982 - 104.83) \text{ kJ/kg} = (6 \text{ kg/min})[(100.88 - 303.87) \text{ kJ/kg}]$$

$$\dot{m}_w = \mathbf{29.1 \text{ kg/min}}$$

(*b*) To determine the heat transfer from the refrigerant to the water, we have to choose a control volume whose boundary lies on the path of heat transfer. We can choose the volume occupied by either fluid as our control volume. For no particular reason, we choose the volume occupied by the water. All the assumptions stated earlier apply, except that the heat transfer is no longer zero. Then assuming heat to be transferred to water, the energy balance for this single-stream steady-flow system reduces to

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{dE_{system}/dt}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} = 0$$

0 (steady)

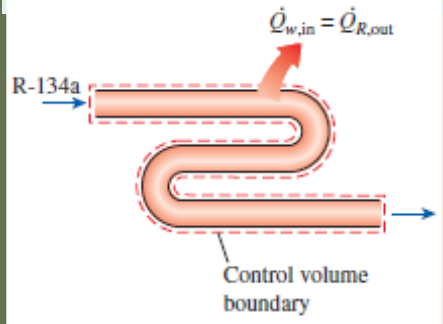


FIGURE 5–41
In a heat exchanger, the heat transfer depends on the choice of the control volume.

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{Q}_{w,in} + \dot{m}_w h_1 = \dot{m}_w h_2$$

Rearranging and substituting,

$$\dot{Q}_{w,in} = \dot{m}_w(h_2 - h_1) = (29.1 \text{ kg/min})[(104.83 - 62.982) \text{ kJ/kg}]$$

$$= \mathbf{1218 \text{ kJ/min}}$$

Discussion Had we chosen the volume occupied by the refrigerant as the control volume (Fig. 5–41), we would have obtained the same result for $\dot{Q}_{R,out}$ since the heat gained by the water is equal to the heat lost by the refrigerant.

5.4 Some Steady-Flow Engineering Devices (9)

5. Pipe and duct flow

- The transport of liquids or gases in pipes and ducts is of great importance in many engineering applications.
- Flow through a pipe or a duct usually satisfies the steady-flow conditions.
- The control volume can be selected to coincide with the interior surfaces of the portion of the pipe or the duct that we are interested in analyzing.
- Under normal operating conditions, the amount of heat gained or lost by the fluid may be very significant, particularly if the pipe or duct is long
- If heat transfer is undesirable, the pipes or ducts are usually insulated, particularly when the temperature difference between the flowing fluid and the surroundings is large.
- Heat transfer in this case is negligible.

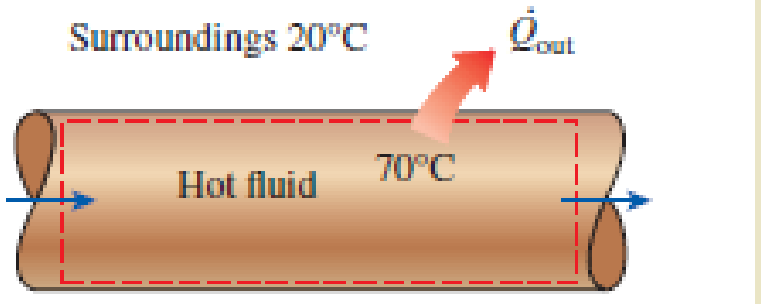


FIGURE 5–42
Heat losses from a hot fluid flowing through an uninsulated pipe or duct to the cooler environment may be very significant.

5.4 Some Steady-Flow Engineering Devices (10)

- Sometimes heat transfer is desirable and is the sole purpose of the flow.
  - Water flow through the pipes in the furnace of a power plant.
  - the flow of refrigerant in a freezer.
  - the flow in heat exchangers.
- If the control volume involves a heating section (electric wires), a fan, or a pump (shaft), the work interactions should be considered
- The velocities are relatively low, and changes are usually insignificant when the pipe or duct diameter is constant (ke ≅ 0).
- Kinetic energy changes may be significant for gas flow in ducts with variable cross-sectional areas especially when the compressibility effects are significant.

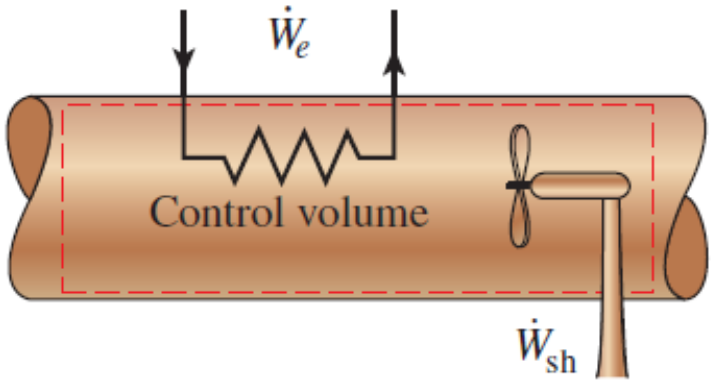


FIGURE 5–43
Pipe or duct flow may involve more than one form of work at the same time.

EXAMPLE 5–11 Electric Heating of Air in a House

The electric heating systems used in many houses consist of a simple duct with resistance heaters. Air is heated as it flows over resistance wires. Consider a 15-kW electric heating system. Air enters the heating section at 100 kPa and 17°C with a volume flow rate of 150 m³/min. If heat is lost from the air in the duct to the surroundings at a rate of 200 W, determine the exit temperature of air.

SOLUTION The electric heating system of a house is considered. For specified electric power consumption and air flow rate, the air exit temperature is to be determined.

Assumptions **1** This is a steady-flow process since there is no change with time at any point and thus $\Delta m_{cv} = 0$ and $\Delta E_{cv} = 0$. **2** Air is an ideal gas since it is at a high temperature and low pressure relative to its critical-point values. **3** The kinetic and potential energy changes are negligible, $\Delta ke \cong \Delta pe \cong 0$. **4** Constant specific heats at room temperature can be used for air.

Analysis We take the *heating section portion of the duct* as the system (Fig. 5–44). This is a *control volume* since mass crosses the system boundary during the process. We observe that there is only one inlet and one exit and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. Also, heat is lost from the system and electrical work is supplied to the system.

At temperatures encountered in heating and air-conditioning applications, Δh can be replaced by $c_p\,\Delta T$ where $c_p = 1.005$ kJ/kg·°C—the value at room temperature—with negligible error (Fig. 5–45). Then the energy balance for this steady-flow system can be expressed in the rate form as

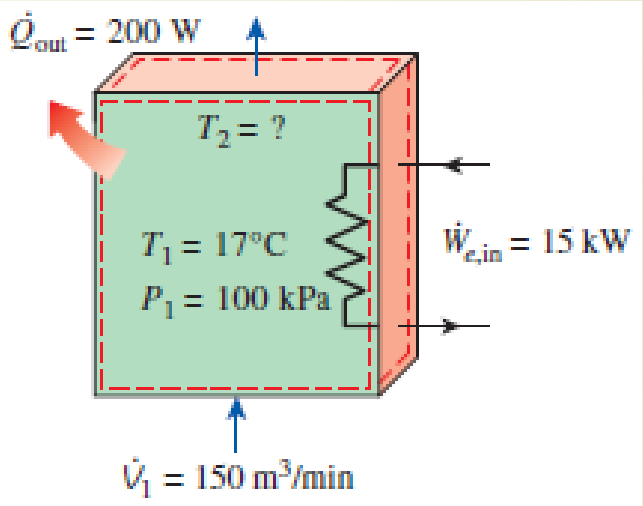


FIGURE 5–44
Schematic for Example 5–11.

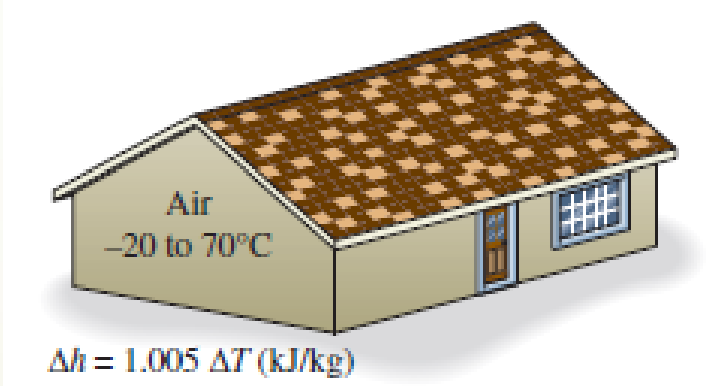


FIGURE 5–45
The error involved in $\Delta h = c_p\,\Delta T$, where $c_p = 1.005$ kJ/kg·°C, is less than 0.5 percent for air in the temperature range −20 to 70°C.

$$\underbrace{\dot{E}_{in} - \dot{E}_{out}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\overset{0\ (\text{steady})}{dE_{system}/dt}}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} = 0$$

$$\dot{E}_{in} = \dot{E}_{out}$$

$$\dot{W}_{e,in} + \dot{m}h_1 = \dot{Q}_{out} + \dot{m}h_2 \quad (\text{since } \Delta ke \cong \Delta pe \cong 0)$$

$$\dot{W}_{e,in} - \dot{Q}_{out} = \dot{m}c_p(T_2 - T_1)$$

From the ideal-gas relation, the specific volume of air at the inlet of the duct is

$$v_1 = \frac{RT_1}{P_1} = \frac{(0.287\ \text{kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(290\ \text{K})}{100\ \text{kPa}} = 0.832\ \text{m}^3/\text{kg}$$

The mass flow rate of the air through the duct is determined from

$$\dot{m} = \frac{\dot{V}_1}{v_1} = \frac{150\ \text{m}^3/\text{min}}{0.832\ \text{m}^3/\text{kg}}\left(\frac{1\ \text{min}}{60\ \text{s}}\right) = 3.0\ \text{kg/s}$$

Substituting the known quantities, the exit temperature of the air is determined to be

$$(15\ \text{kJ/s}) - (0.2\ \text{kJ/s}) = (3\ \text{kg/s})(1.005\ \text{kJ/kg}\cdot{}^\circ\text{C})(T_2 - 17)^\circ\text{C}$$

$$T_2 = \mathbf{21.9^\circ C}$$

Discussion Note that heat loss from the duct reduces the exit temperature of air.

Exercise 3

5–44 Refrigerant-134a enters an adiabatic compressor as saturated vapor at −24°C and leaves at 0.8 MPa and 60°C. The mass flow rate of the refrigerant is 1.2 kg/s. Determine (*a*) the power input to the compressor and (*b*) the volume flow rate of the refrigerant at the compressor inlet.

Exercise 4

5–46 Steam flows steadily through an adiabatic turbine. The inlet conditions of the steam are 4 MPa, 500°C, and 80 m/s, and the exit conditions are 30 kPa, 92 percent quality, and 50 m/s. The mass flow rate of the steam is 12 kg/s. Determine (*a*) the change in kinetic energy, (*b*) the power output, and (*c*) the turbine inlet area. *Answers:* (*a*) −1.95 kJ/kg, (*b*) 12.1 MW, (*c*) 0.0130 m^2

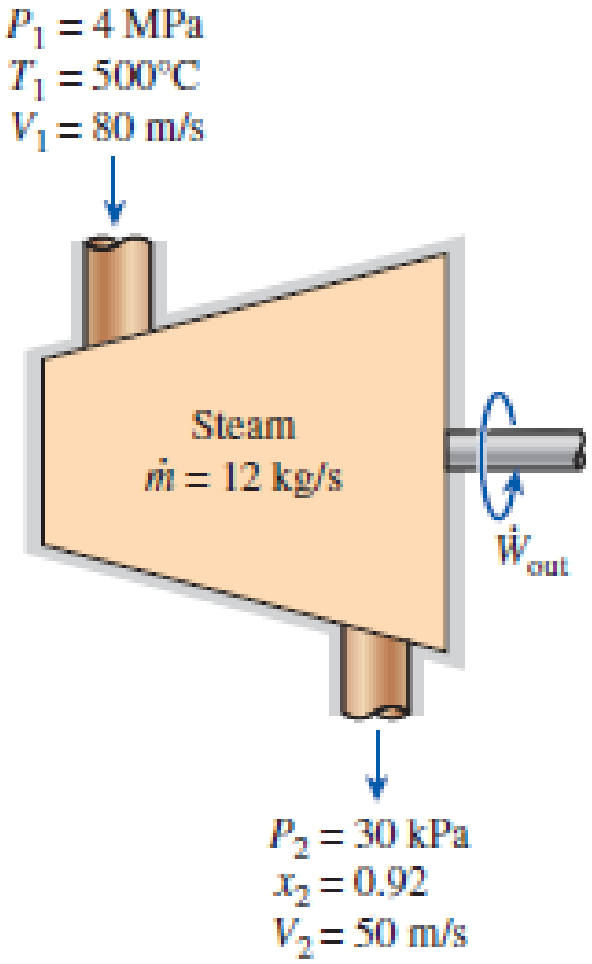


FIGURE P5–46

Exercise 5

5–54 An adiabatic gas turbine expands air at 1300 kPa and 500°C to 100 kPa and 127°C. Air enters the turbine through a 0.2-m^2 opening with an average velocity of 40 m/s, and exhausts through a 1-m^2 opening. Determine (*a*) the mass flow rate of air through the turbine and (*b*) the power produced by the turbine. *Answers:* (*a*) 46.9 kg/s, (*b*) 18.3 MW

Exercise 6

5–72 In steam power plants, open feedwater heaters are frequently utilized to heat the feedwater by mixing it with steam bled off the turbine at some intermediate stage. Consider an open feedwater heater that operates at a pressure of 1000 kPa. Feedwater at 50°C and 1000 kPa is to be heated with superheated steam at 200°C and 1000 kPa. In an ideal feedwater heater, the mixture leaves the heater as saturated liquid at the feedwater pressure. Determine the ratio of the mass flow rates of the feedwater and the superheated vapor for this case. *Answer:* 3.73

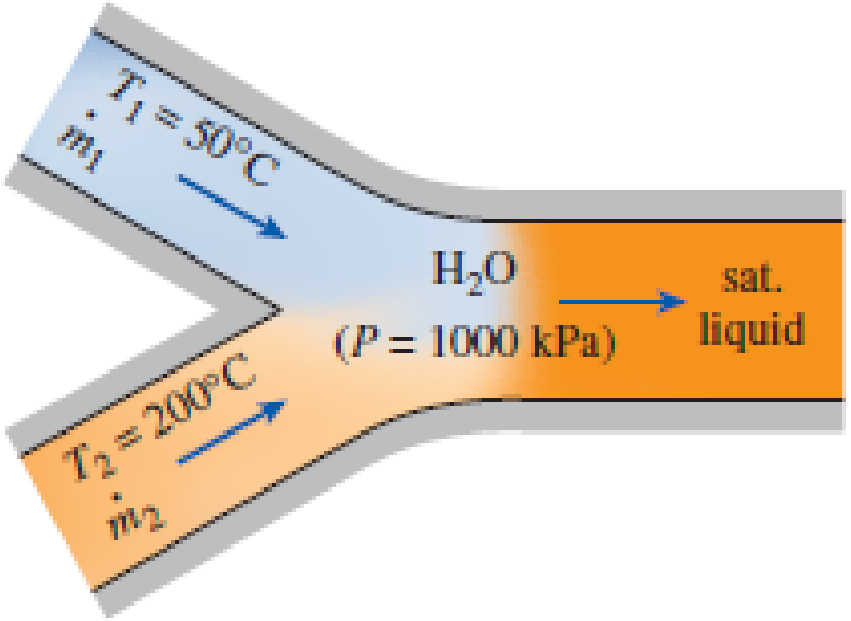


FIGURE P5–72

Exercise 7

5–105 Refrigerant-134a enters the condenser of a refrigerator at 900 kPa and 60°C, and leaves as a saturated liquid at the same pressure. Determine the heat transfer from the refrigerant per unit mass.

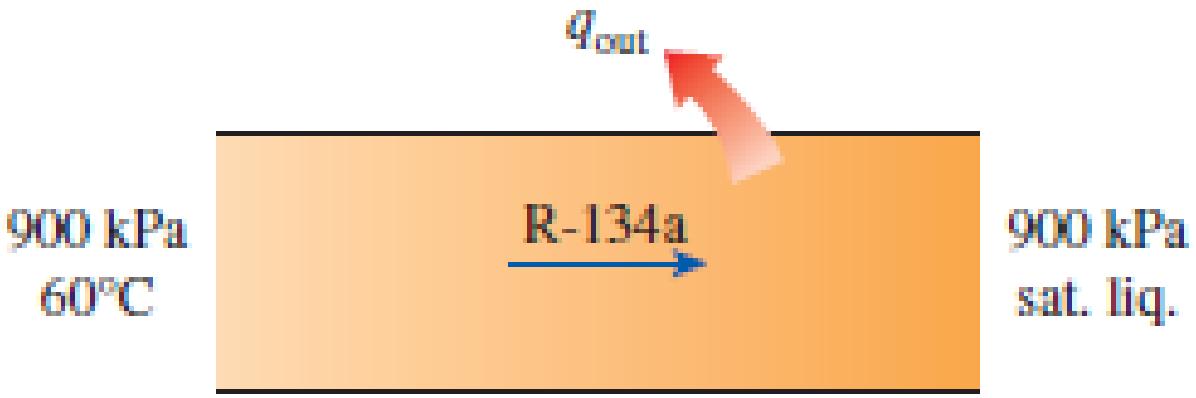


FIGURE P5–105

Thank You